



MECH MAINT- OFF SITE Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2025-26/091034/397541

Date:

26/08/2025(16:11:57)

Subject:APPROVAL FOR LMI ON GUIDE LINES ON CW DUCT
INSPECTION COS -ISO-00-OGNOPSMECH048

Approval for LMI on GUIDE LINES ON CW DUCT INSPECTION, COS -ISO-00-
OGN/OPS/MECH/048

Approval for GUIDE LINES FOR CW DUCT INSPECTION, COS -ISO-00-OGN/OPS/MECH/048 is due
for revision.

It is requested to approve the same.

**ISSUE NO.3; REVISION NO. 2.00; DATE: August 25, GUIDE LINES FOR CW DUCT
INSPECTION, with ref to COS -ISO-00-OGN/OPS/MECH/048 is submitted for approval &
implimentation please.**

Submitted by:

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Note No:2

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Note No:3

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

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Note No:4

Modified as per standard format of Simhadri LMI and attched .

May please process further.

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Note No:5

It is to be put up in standard format

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Note No:6

Modified as per standard format .

May please process.

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Note No:7

Reviewed

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Note No:8

LMI is in line with the OGN. Forwarded for further review.

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Note No:9

Checked and forwarded.

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Note No:10

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Note No:11

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Note No:12

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Note No:13

Reviewed. File may be processed further.

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Note No:14

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Note No:15

LMI is in line with the OGN. May please be approved.

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On:01/10/2025(10:20:03) **Ref:**SMTTPP/MECH MAINT-/2025-26/091034/397541

Note No:16

May please approve.

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Note No:17

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Note No:18

Approved & Submitted by:

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On:03/10/2025(20:09:05) **Ref:**SMTTPP/MECH MAINT-/2025-26/091034/397541

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

GUIDELINES FOR CW DUCT INSPECTION

LMI/OGN/OPS/MECH/048

AUG -2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	GUIDELINES FOR CW DUCT INSPECTION
2	LMI No.	LMI/SMSTPS/OGN/OPS/MECH/048 Rev 2
3	Reference Document	COS-ISO-00-OGN/OPS/MECH/048
4	Superseded Document	OGN-OPS- MECH-048: Rev 1, Aug 2023
5	Prepared By	Name: R. Murali Mohana Rao Sr Mgr. (MM-OFS)
6	Checked By	Name: T.V Rajasekar AGM (MM-OFS)
7	Reviewed By	Name: Hemendra Kumar Verma GM(MM)
8	Reviewed By	Name: Ahmad Kaunain Raza GM(O&M)
9	Approved By	Name: Sameer Sharma HOP

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1.0 Introduction

CW Duct of circulating water system is an important link to take the condenser hot water

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to Cooling Tower for its cooling. This is partly over ground and mostly underground / buried. Thus, periodic inspection of this duct becomes imperative and of utmost requirement.

Similarly, healthiness of Air Release Valves (ARVs) is also required apart from checking the large size butterfly valve of circulating water system duct. Inspection and Repair of PU coating is very essential to have longer life of the duct. Since this is mostly underground & buried, sufficient and required safety precautions need to be taken before entering into the duct.

2.0 Superseded Documents

OGN-OPS- MECH-048: Rev 1, Aug 2023

3.0 Details of Revision

Rev-2: The LMI document is redrafted as per standard LMI format

4.0 Identification of LMI

LMI/SMSTPS/OGN/OPS/MECH/048. Rev -2.

5. Guidelines

5.1 Pre inspection requirements

- Ensure CW duct layout drawings are available for understanding the actual route of the pipeline/duct right from CW pump house to Condenser and from Condenser outlet to Cooling Tower inlet.
- Identification of the Air Release Valves (ARV) and their location should be done.
- Ensure availability of the protocol for joint inspection of the duct before first charging and check for any unwanted comment on the same.
- There should not be any temporary or permanent construction / growth of trees / erection of electric poles / telephone lines on the buried route of the CW duct / pipeline.

5.2 External Inspection

- Ensure that the ARVs are always kept in open condition.
- To have a walk down inspection of the duct / pipeline route for any

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leakage / seepage before inspection.

- To ensure that the pits made for Pitot tube insertion for flow measurement are dry and proper platform with ladder / stair are available. The pit should be properly covered so that no man, material, animal fall in the pit.
- To check & ensure that the buried duct / pipeline is properly coated / encased to prevent from external corrosion.
- The exposed duct / pipeline needs to be painted as per guidelines on regular interval / during each unit overhaul.
- Healthiness and smooth operation of Interconnection, blow down valves and debris filters inlet valves

5.3 Precautions during Internal Inspection

The internal inspection of the CW duct / pipeline is required to be done during unit overhaul with all precautions as given below

PTW for CW duct inspection is required to be taken. PTW for CW duct inspection to be issued only when Oxygen level is between 19.5% to 23.5%. Cross permit for isolation of CW pumps needs to be ensured with an instruction that the CW pumps should be run only when duct inspect PTW has been returned with declaration of removal of man & materials.

Isolation of CW pump discharge valves, duct Interconnection valves, blow down valves, condenser inlet & outlet valves are to be kept closed electrically / hydraulically & locked manually.

5.4 Precautions during Internal Inspection

- All the ARVs of cold as well as hot water ducts, CT Inlet butterfly valve, CT Riser valves should be kept opened for air entry & air out
- The duct / pipeline needs to be completely emptied / drained with the help of sufficiently large size dewatering pumps and water should be thrown far from the routing of the pipeline so that in case of leakage / seepage in the pipeline, there is no recirculation of water inside the duct.
- Left over water in the duct shall be removed by dewatering pumps and cleaning & scooping. It should be ensured that cables supplying power to pumps are in healthy condition and kept isolated from duct surface through suitable means.

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- All the manhole covers need to be removed from position for entry of inspecting team and ventilation purpose.
- Proper ventilation arrangement needs to be made available before entering the duct so that sufficient air flow is established inside the duct with the help of large size ventilation fans. Oxygen level needs to be measured and checked that it is safe to enter.
- DC lights and walkie-Talkie of sufficient range need to be made available to the inspecting team of healthy persons during internal inspection.
- Another team with all the PPEs, torch lights, first aid box etc. shall be available outside, during the entire inspection period and shall be in contact with the inside team for any rescue
- / help.
- Proper PPEs such as masks, gum boots, helmet, 24 V DC portable search lights, sticks, ropes etc. should be with the team inspecting the duct. Fall Arresters to be used while climbing up and down the rung ladders.

5.5 Internal Inspection

- The inside of the duct needs to be thoroughly inspected for corrosion, rusting, pitting, dents, any abnormality in the circularity of the pipeline, debris, scraps, unwanted materials etc. should be noticed and proper distance of the defective area / defective point should be mentioned in the joint inspection report for further rectification measure. The distance recorded should be with respect to the entry point or end point or ARV location.
- After carrying out the inspection, a joint inspection report shall be made by the team members and submitted to the head of operation, head of maintenance & head of O&M for their information and necessary action.

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6. Distribution

(through e-distribution)

- GM (O&M)
- Head of Maintenance
- Head of Operation
- Head of EEMG
- Head of EMD
- Head of C&I
- Head of MM/CPC
- Head of Chemistry
- Head of Civil
- Head of FQA

7.Review

HOD (OFS) will review & take the approval of competent authority (HOP).
The document will be reviewed on 2 yearly basis or earlier, if required.

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8. Checklist

8.0 For CW internal duct inspection for sea water station

Sl No	Check	Responsibility	Frequency	Remarks (Yes/No)
1	Identification of all Manholes in Supply duct & return duct and Air release Valves of the unit	MM/OFS	Overhaul	
2	Ensure CW duct draining by dewatering	MM/OFS	Overhaul	
3	Ensure Proper isolation of CW Pump Discharge Valves, Interconnection Valves & Blow down valves	Operation & MM/OFS	Overhaul	
4	Ensure proper hard barricading around the manhole area	MM/OFS	Overhaul	
5	Ensure Oxygen level inside the duct is between 19.5% - 23.5%	Chemistry	Overhaul	
6	Ensure PTW, Work at Height & Confined space permit while going inside the duct	Operation, Chemistry & MM/OFS	Overhaul	
7	Ensure dehumidifiers in CW duct in case of high humid conditions	MM/OFS	Overhaul	
8	Ensure proper ventilation of ducts	MM/OFS	Overhaul	
9	Assess the condition of Internal coating inside the duct	MM/OFS	Overhaul	

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SI No	Check	Responsibility	Frequency	Remarks (Yes/No)
10	Ensure hot work permit in case of any welding if required	MM/OFS	Overhaul	
11	Surface of preparation internal duct surface	FQA (Yes/No) MM/OFS	Overhaul	
12	Quality checks of the internal coatings done	FQA & MM/OFS	Overhaul	
13	ARV's isolation valve servicing	MM/OFS	Overhaul	
14	ARV's checking including its mounting stub health assessment: Replace balls & seals if found damaged	MM/OFS	Overhaul	
15	Annubar inspection to be done visually, followed by DPT examination to identify any cracks or wear on sensor.	C&I	Overhaul	
16	Ensuring DC supply lights inside the duct	EMD	Overhaul	
17	Ensuring duct is free from all foreign materials & all living beings prior to charging	Operation & MM/OFS	Overhaul	
18	Pitot tube arrangement and pit healthiness to be checked	MM/OFS	Overhaul	
19	Ensure all the gaskets are replaced for Manholes, damaged fasteners replacement & box up	Operation & MM/OFS	Overhaul	

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8.1 For CW external duct inspection for sea water station

SI No	Check	Responsibility	Frequency	Remarks (Yes/No)
1	Visual inspection for any seepages.	MM/OFS	Overhaul	
2	Repair of any leaks	MM/OFS	Overhaul	
3	Thickness measurement	MM/OFS & FQA	Overhaul	
4	Replacement of pipe/duct if thickness reduces by 40%	MM/OFS	Overhaul	
5	NDT of new weld joints – 100% Root DPT	FQA	Overhaul	
6	Painting Check	MM/CPC	Overhaul	
7	Concrete duct repair	Civil & MM/OFS	Overhaul	

9. Annexure

SI No	Record Description	Maintained by	Frequency	Remarks
1	CW Internal Duct inspection for Sea water Stations	MM/OFS	Overhaul	
2	CW External duct inspection	MM/OFS	Overhaul	